



The management of cross-functional inter-dependencies in ERP implementations: emergent coordination patterns

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Abstract

This paper reports a study examining cross-functional coordination in enterprise resource planning (ERP) projects. We collected primary data related to ERP implementations at four large organizations representing different industries and implementation strategies. These are presented through rich descriptions organized on the basis of an emergent process theory framework. Our case studies suggest that there are three main patterns of managing functional inter-dependencies:

- (a) A lean coordination pattern that involves intricately planned 'vanilla' implementations using reference process models and strong methodologies
- (b) A rich coordination pattern that is based on managing inter-dependencies through organizing arrangements and cultural interventions, and
- (c) A mediation pattern that is based on executive mandate or a dominant functional unit laying out the rules of engagement.

This study provides practical implications for industry managers engaged in large-scale enterprise system implementations. It also significantly advances theory by outlining and characterizing cross-functional coordination patterns. Our findings confirm that studying surface regularities as well as the underlying situated phenomena, in combination, can help in generating new insights for guiding ERP projects.

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It's like squeezing a balloon...you make a change in one place and see what pops out at another place...you change a thing in manufacturing...something pops out in A/R
[Project Manager at ConsCo]

Introduction

Enterprise resource planning (ERP) systems are intended to enable the integration of transaction-oriented data and business processes throughout an organization (Markus & Tanis, 2000). A core benefit promised by these systems is the seamless integration of all information flowing through the company – financial and accounting information, human resource information, supply chain information, and customer information (Davenport, 1998). The goal of enterprise integration demands co-adaptation of different functional areas in an organization into a new process framework enabled by an integrated Information Technology (IT) platform. As increasing numbers of organizations across the globe have

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chosen to deploy ERP applications, there has been a greater appreciation for the challenges involved in implementing these complex technologies (Themistocleous *et al.*, 2001; Robey *et al.*, 2002; Olhager & Selldin, 2003). In the face of these problems, a number of studies have reported on factors associated with implementation success such as top management support and change management (Nah *et al.*, 2001). Qualitative studies, however, suggest that there is a gap between empirical validation of those factors and theoretical expectations (Sarker & Lee, 2000; Bagchi *et al.*, 2003). For example, factors such as user involvement may have different connotations in different contexts (Bagchi *et al.*, 2003) while others such as balanced and empowered team may not be a necessary condition for successful implementation (Sarker & Lee, 2000). Besides, despite an abundance of empirical studies, little progress has been made in generating focused theories that can provide an understanding of how these factors inter-relate and affect overall success (Nah *et al.*, 2001).

Akkermans & van Helden (2002, p. 44) propose that 'the core process on any successful implementation consists of mutually reinforcing communication and collaboration between project team members from different departments and business functions'. This study proposes to better understand this core-process, which we label cross-functional coordination.

Coordination is a process of linking together different activities of organizations to accomplish a common goal (Van de Ven *et al.*, 1976). The linking together of different activities in organizations requires the management of dependencies (Malone & Crowston, 1990). *Cross-functional coordination* refers to the need for resolution of dependencies among different business function areas in an Information Systems (IS) project. The focus of this paper is specifically on the implementation phase of ERP projects – to explore how different business functions work in concert to make sure that the implemented system supports cross-functional business processes.

Cross-functional coordination is important in any IS project (Kraut & Streeter, 1995), but as the quote at the start of this paper suggests, it is especially critical in ERP projects due to the nature of integrated functionality and a broad scope that affects all functional areas. The goal of enterprise integration demands change from different functional areas, breaking functional silos, and re-orienting around value-creation processes enabled by the integrated IT platform. The very process by which these systems are put in place also requires coordination across the enterprise – in arriving at a shared vision of end-to-end processes, making joint resource commitments, sharing information, and working together for enterprise goals. Therefore, the process of coordination among functional units may require significant effort that outweighs the resources needed for up-front software purchase (Scheer & Habermann, 2000). The problem of coordination is counted as one of the most important

issues leading to failure of a number of ERP implementations (Kumar & Van Hillegersberg, 2000).

This paper reports on a study employing the case study methodology (Eisenhardt, 1989) and the functional coordination perspective to generate theory to inform ERP implementations. In our methodological approach we affirm a reciprocal relationship between data collection, analysis, and theory. We started out with a delineated area of study together with an 'orienting framework', and allowed relevant theoretical insights to emerge. Our analytical approach was configurational in nature – we are concerned with how patterns of multiple independent variables are related to dependent variables rather than with how individual independent variables affect the dependent variable (Doty *et al.*, 1993).

We pose the following research questions:

- What patterns of cross-functional coordination emerge over the course of implementation in different organizations and what contingency conditions shape them?
- What is the role of cross-functional coordination in the ERP implementation process and how does it affect the overall success of the implementation?

The rest of the paper is organized as follows. In the next section, we develop a theoretical framework that conceptualizes emergent patterns of cross-functional coordination in ERP implementation as a contingency fit between coordination needs and organizing elements. This is followed by a description of the methodology, and further sections present the findings based on case studies of four ERP implementations a discussion of the findings. The last section concludes the paper.

Cross-functional coordination

Drivers, organizing elements, and emergent patterns

An orienting framework (e.g., Hackman, 1990) helps to guide and focus data collection and analysis in the investigation of a phenomenon. We base this study upon the emergent process theory framework proposed by Soh & Markus (1995). The framework suggests that goal directed behavior interacts with environmental conditions, resulting in outcomes that may be different than intended. Emergent process theory emphasizes the unpredictable interactions between people in organizations and the environment. Soh & Markus (1995) frame the 'IT investment to business value' process as a series of linked models that roughly correspond to the phases of a typical IT investment. The outcomes of each phase result from the interactions between uncontrollable events and choiceful human actions, as ERP implementations are particularly susceptible to unintended side-effects (Hanseth *et al.*, 2001). Following the framework, we study both the situational factors that create a need for coordination, as well as organizing elements that facilitate functional coordination (Figure 1). We note uncontrollable events that lead to emergent outcomes different from the intended consequences.

Our orienting framework posits a contingency fit between the situational factors creating need for cross-functional coordination and the organizing elements facilitating coordination. The match between the two indicates effective coordination and the type of fit indicates different archetypal coordination patterns. The need for cross-functional coordination arises due to *complexity* (size and scope of the project and the functional complexity of the organization), *inter-dependencies* in work activities across functional areas, and *uncertainty* in the organization's environment. Organiza-

tions deal with the need for coordination by enacting *structures*, *processes*, and *communication* to better exchange and respond to information. The need for a match between situational factors and organizational configurations has been long established in organizational (Mintzberg, 1979) and information systems (Chan *et al.*, 1997) literature. Recently, Barki & Pinsonneault (2005) have proposed that a similar contingency fit applies to ERP implementations.

Emergent patterns of coordination

March & Simon (1958) propose that coordination may be based on pre-established schedules – *coordination by plan*, and coordination that involves transmission of new information – *coordination by feedback*. Mintzberg (1983) echoes a similar typology in suggesting that the three different ways of approaching coordination are standardization, mutual adjustments, and direct supervision. Table 1 summarizes other studies that map to these three types of coordination patterns.

Coordination through mutual adjustment is based on rich information that allows for the inter-dependencies to be communicated and adaptation responses to be negotiated. In ERP projects, mutual adjustment may be supported through different organizing elements. Structurally, mutual adjustment may be supported through the design of cross-functional teams – this would allow for functional requirements to be voiced in the project team and for suitable strategies supporting different

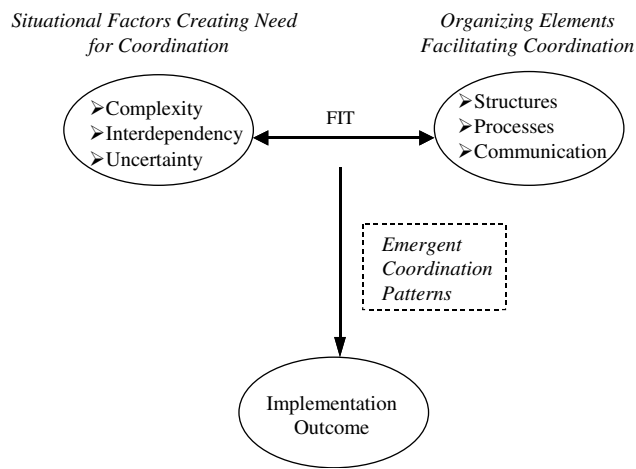


Figure 1 Orienting framework.

Table 1 Coordination typology derived from extant theory

	Coordination by mutual adjustment	Coordination by mediation	Coordination through standardization
March & Simon (1958) Lawrence & Lorsch (1967)	Coordination by feedback	Integrative subsystems, specialists in integration, integrating teams	Coordination by plan
Van de Ven <i>et al.</i> (1976)	<i>Personal mode</i> and <i>group mode</i> (mutual task adjustments through vertical or horizontal channels of communication)		<i>Impersonal mode</i> – pre-established plans, schedules, formalized rules, policies and procedures, and standardized information and communication systems
Galbraith (1973)	Creation of slack resources, self-contained task groups, information systems, lateral linkages		Rules, programs and procedures, common understanding of targets and goals
Daft & Lengel (1986)	Direct contact, group meetings	Integrators	Plans, formal information systems, rules and regulations
Mintzberg (1983)	Mutual adjustment	Direct supervision	Standardization of work processes, outputs, skills and knowledge, norms
Crowston (1991)			Interdependencies between tasks, interdependencies between task and object, interdependencies between objects
Gittell (2002)	Relational coordination	Boundary spanners	Routines

Table 2 Case studies

Context	Enterprise	Key respondents
Major telecom equipment manufacturer	TelEquipCo	IT-director, project manager (2)
Diversified consumer goods manufacturer	ConsCo	IT-director, project manager (2)
Regional electrical power utility	UtilityCo	CIO, IT-director, project manager (3)
Global Financial Institution	GlobalFinCo	CIO, project manager, functional area liaisons (7)

functional areas to be negotiated. In terms of communication and information exchange design, mutual adjustment would be supported by supporting open communication among functional units, communications between functional units and project teams, and communications among project teams. In terms of process design, mutual adjustment would be supported by modularly designed functional tasks where the locus of activities and control for each functional area is clearly mapped and understood.

Coordination through direct supervision or mediation is a way to achieve coordination by having an organization sub-unit or a role take on the responsibility for the conflict resolution. Structural ways to achieve such coordination could be a steering committee consisting of senior management from different functional areas that takes on responsibility for resolving conflicts (Slater, 1998). Other means of mediation may devolve the responsibilities for major decisions related to the ERP implementation on a competent and neutral group or a functional area that spearheads the initiative. From a process design perspective *ad hoc* top management involvement in different phases of the implementation may help to achieve coordination and break deadlock over major issues.

Coordination through standardization may occur in three different ways: standardization of work-processes, standardization of work outputs, and standardization of inputs such as worker-skills. In ERP projects, the standardization of work processes is realized by adopting a comprehensive project methodology that provides a shared template for the implementation to all functional areas. The standardization of work-skills may be realized by training and socialization. Finally, the standardization of work outputs refers to the outcomes of the implementation in terms of the instantiated systems and IT-embedded processes. The use of reference business models allows for a structured way of coordination through standardization of these outcomes.

Methodology

This study consists of four case studies of ERP implementations at four large organizations (Table 2). Large organizations were chosen as they are expected to have more complex coordination dynamics and face greater institutional pressures resisting change (Christiaanse & Huigen, 1997). While the four companies are headquartered in U.S.A., they show a rich diversity in their business conditions, implementation choices, and con-

tingencies affecting cross-functional coordination – in line with recommendations for case-based research (Eisenhardt, 1989; Yin, 1994). The cases were developed based on retrospective recollection of the implementation process and associated conditions by senior IT and business executives. The structured interviews were conducted across multiple sessions at the company headquarters located in the United States. We allowed a coherent narrative¹ to emerge from the interviews that organized events and organizational actions into a whole that ‘gives meaning to and explains each of its elements and is, at the same time, constituted by them’ (Griffin, 1993, p. 1097). In addition, we collected and organized news articles about the ERP implementations by these companies. To familiarize us with the company context during the period of the study, we also examined materials at the company websites and in secondary databases.

The interview protocol consisted of a set of questions dealing with each of the project stages (see Appendix). The interview questions were open-ended and the answers were transcribed separately by each of the three authors. The interviewees were not made aware of the main research questions to prevent possible biases. The protocol in particular helped the respondents weave a ‘storyline’ for the entire implementation process and provide a higher-level narrative into which the specific details could be appropriately placed. We also assessed the overall success of the project as perceived by the highest-ranking functional executives. This was independently confirmed through company news announcements and stories in the business press.

Purposive and theoretical sampling was used to allow for diversity in the nature of cross-functional coordination – the organizations differ in their coordination needs and organizing elements. Theory was allowed to emerge from the data through a process of constant comparison to check for differences across implementation contexts and in searching for possible explanations. The three authors independently took notes during interviews and used memos to identify coordination needs, provision, and emergent patterns. Following the

¹Attention to coherence focused on ensuring that we had a complete timeline of the project and the events fit the timeline; assessments of different executives were consistent among themselves and with external news stories and reports; and, attributions about events and actions and the implied causal reasoning were mutually consistent.

conclusion of all data gathering for a particular case, each author shared notes and memos with the others and these were then discussed and a consensus developed. A detailed case write-up was then prepared and situational factors and organizing elements were assessed. Table 3 summarizes how our research methodology met appropriate criteria to ensure validity of our analysis. The definitions of validity tests and tactics are based on Yin (1994). The implementation followed previous exemplary studies using case study research methodology (Rindova & Kotha, 2001).

Findings

Case study 1: TelEquipCo – a contentious package selection

TelEquipCo perceived the ERP implementation as an opportunity to get away from legacy systems that were mostly oriented towards a regulated environment. Existing systems were kludged together and made enterprise-wide information sharing difficult. However, there were clear disagreements among the functional areas on the specific ERP package to adopt. At the project's initiation, about 25% of the organization was running some version of Oracle applications. Also, a large contract in Saudi Arabia had been supported with a standalone Oracle ERP installation that had positive results in terms of speed of

implementation, operating performance, and 'happiness' of end-users.

At this time, the CIO unilaterally decided to go with SAP and also convinced the board for SAP. However, the decision did not have a substantial buy-in from the business side, and was made more on 'gut-feel' rather than a sound study of company's business processes. Overall, the lack of a business buy-in prevented much of the value of the system from being realized.

Situational factors

Complexity: TelEquipCo designs, develops and manufactures communications systems, software and products. One of the main complicating factors at the company was the dynamic nature of the industry that meant constantly changing company structure and new acquisitions. The company's acquisition of WestCorp, for example, created a clash of business models with TelEquipCo's legacy of slow and steady growth, at contrast with the enterprising 'silicon valley' culture of WestCorp.

Inter-dependency and functional departmentalism: The company was organized as about 433 legal entities with dispersed corporate functions. The management model was complex with project, legal, product, and market-based entities. TelEquipCo's complicated management models of 'constructive contention' between

Table 3 Ensuring different types of validity

<i>Tests</i>	<i>Case study tactics</i>	<i>Implementation in this study</i>
Construct validity Establishing correct operational measures for the concepts being studied	1. Use of multiple sources of evidence 2. Establish chain of evidence Have key informants review draft of case study report	Multiple source of evidence based on key respondent interviews supplemented by study of press releases and media coverage Consistent procedures reported in the paper used for framing constructs related to coordination Cases for each company presented back to executives
Internal validity Establishing causal relationships whereby certain conditions are shown to lead to other conditions	1. Pattern matching: compare observed relationships to theoretically expected ones 2. Explanation building: stipulate causal explanations 3. Time series analysis: compare unfolding process across time	Findings confirmed theoretical predictions indicating that firms match coordination needs with contingent organizing elements facilitating coordination Observed coordination patterns matched typologies suggested in past literature Narrative constructed based on respondent interviews allowed understanding of processes unfolding over time
External validity Establishing the domain to which findings can be generalized	Use replication logic in multiple case studies	Four firms that implemented ERP packages were selected
Reliability Demonstrating that the operations of the study – such as data collection procedures – can be repeated, with the same results	1. Develop case study database 2. Use case study protocol	Data were organized in a case database with a chronological list of actions/events Interview protocol was laid out and consistently followed Multiple authors independently interpreted the patterns of coordination and an independent expert read through the cases to verify the interpretation of coordination pattern

different business units meant relatively independent functional units.

Uncertainty: The telecommunications equipment industry has been characterized by a high rate of technological uncertainty. There have been great competitive pressures brought upon by the shift from circuit switching to packet switching technologies and the rise of new challengers that come from, hitherto different but now converging, industries. The response of players like TelEquipCo has been to seek mergers with more nimble players that have competence in the new technologies.

Organizing elements Three distinct phases may be discerned in TelEquipCo's ERP implementation – an exploration phase, a backend financial processes implementation phase, and a front-end quote-to-cash implementation phase. The initial scope of the project was fairly limited. After the initial exploration phase, it was discovered that the SAP initiative was not getting anything done. As a result, the CIO was fired and a new SAP initiative was announced (the 'breakdown').

Organizational structure: At the beginning of the project in 1995, each function made an independent decision on consultant support, leading to a number of different methodologies that did not mesh together. At the time, TelEquipCo was using the services of 6–7 consulting companies. Later, after the breakdown, there was a conscious attempt to enforce standards for codes, notation, and documentation. For the implementation teams, TelEquipCo followed a practice of getting people from the business units, training them and then sending them back to the business units. This helped to put some of the expertise back into the businesses. The finance function and the CEO were highly goal-oriented and pushed business to some extent but the functional areas never truly came along in the SAP implementation. As the implementation progressed, a number of legacy systems were decommissioned. One of the observations during the implementation was that due to integrated nature of the package little gains would be realized with piecemeal implementations – 'If you have SAP, you want more. It is like an addictive drug. When it is very integrated it is easy to work with. If you implement small you really hate it'.

Processes: During the implementation, TelEquipCo drew upon SAP's reference models and stuck to them to minimize customization. As an example, the company's inventory valuation procedure was different from the standard and the project team proposed writing a custom posting routine to override SAP, but the management resisted the argument.

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TelEquipCo tried to blend in-house expertise from past implementations with process and systems expertise from consultants.

Communication and information exchange: At TelEquipCo there was a realization that the company faced a bias to do things functionally while ERP packages do not work with that mindset. A group at the company (Global Process Driver Group) attempted to make people mindful of process dependencies with continual reinforcing – 'If you put in an order incorrectly...the revenue books in the wrong place'. In order to communicate the value derived from the new system, TelEquipCo created a flexible data warehouse based on the ERP. This allowed managers get a pulse of the business with simple views and generate insights into issues such as 'why increasing orders do not result in rising profits?' After the breakdown, there was a conscious attempt to enforce standards for codes, notation, and documentation. All interfaces, custom code, and conversions were also assigned to one group for maintenance.

Evaluation of the project After the revised SAP initiative with a new CIO was announced, it was decided to lay down simple but 'stretch' goals for the implementation. The finance function set clear directions for the initiative. In all, 80% of TelEquipCo's legal entities were targeted to have their ledgers in the system by October 1999. The others were primarily held back by regulatory constraints. By October of 1998, support functions such as consolidation and treasury solutions were already up. The SAP implementation was successful in some respects. It halved the number of legacy systems and it simplified business processes. One ramification of process rationalization was the standardization of processes across the globe. In one instance, the company moved some people from Atlanta to Singapore and they were able to start working productively from the outset using the same process and tools. The ERP implementation created new visibility into the operational causes of financial indicators. It eliminated multiple platforms, batching, and handoff requirements. While the upside of the implementation was that it took out some of the cost and complexity of doing business, on the downside it was not the 'two clicks away from every piece of data' case that had been presented to the board. Overall, the project ended two years overdue and at an additional cost of more than \$175 million.

In retrospect, TelEquipCo management came to the realization that while the SAP vs Oracle debate generated much passion around software package choice, in the end that was not as important as making sure the implementation met business needs through a change in business processes and getting committed buy-in from all functional areas.

Overall coordination pattern: High level of conflict over choice of package imposed by CIO. Complex coordination needs and functional conflict led to a breakdown.

Project re-initiated based on attention to common standards.

Case study 2: ConsCo – complex implementation in a turbulent business climate

The choice of SAP for the ERP package was made in late 1994. It was driven by the manufacturing groups several of whom did not have Y2K-compliant systems. Finance had also been considering upgrading its legacy batch system but decided to move to SAP since the package had already been purchased for manufacturing and included a 'free' general ledger. The SAP project was initially intended for U.S. businesses only and later included Europe, but the remaining countries of the world, mostly running J.D. Edwards, were to be left as is. The original schedule called for a manufacturing plant to go live in May 1996 with order management, distribution, and accounts receivable to go live in parallel. The remaining locations/functions were left open, depending on experience with earlier implementations.

Situational factors

Size and complexity: ConsCo had sales of \$6.4 billion and 36,000 employees worldwide. It was organized into three core lines of business: pharmaceuticals, consumer health products, and confectionary products. The complex structure at ConsCo was reflected in the multitude of legal entities. Also it was compounded by changes. In 1994, for example, the consumer health division was spun off into a joint venture with a special set of new reporting requirements. The scope of the ERP implementation was fairly large with manufacturing followed by order management, distribution, and accounts receivables.

Inter-dependency and functional departmentalism: ConsCo had numerous smaller businesses included in its core product lines such as shaving products, tetra fish, products and empty gelatin capsules. All divisions, including the smaller ones, had relative autonomy. However, some functions, such as accounts receivable, accounts payable, and distribution were centralized and profit and loss decisions were subject to corporate control. ConsCo had a complex legal entity structure (over 30 legal entities). Further, the legal entity structure was very different from the management structure – for instance, the three core businesses were one legal entity. One of the key questions ConsCo faced, given its global operations, was whether this should be a global system but that idea was dropped, as it was felt that no one group should maintain standards for global operations.

Uncertainty: ConsCo operated in a fairly turbulent environment and the business was constantly 'patching' in response to competitive market conditions. As an example, the healthcare business was spun into a joint-venture at the last moment before the project started creating a special set of reporting requirements. In the words of ConsCo executives – '*business does not stand still*'

– shortly before the order-management piece was scheduled to go live the generic pharma unit was sold off.

Organizing elements

Organization structure: ConsCo decided to go in with a multi-disciplinary implementation team. The team was located away from day-to-day activities. Initially, it consisted of some full-time and some part time members but eventually it was decided that part time assignment did not work out. The off-site location did help development, but also posed a divide from users entrenched in their own world. The team composition varied but generally included in-house functional experts, in-house IS experts, implementation partner's (PwC) consultants, and third-party consultants (primarily in the Basis team). At its peak, roughly 125 people, with about 40% of them ConsCo employees, were involved in the implementation. The project organization began with the appointment of functional team leaders in core areas – Financial accountin and control (FI-CO) team, Accounts receivables (A/R) team, Sales and distribution (SD) team, and Materials Management (MM) team). A Human Resources (HR) team was originally included but after several months it was determined that the HR module would not provide the expected functionality and the HR team moved to a Peoplesoft implementation. The team leaders were able to negotiate for their staff. In addition, a technical team was staffed with members competent in databases, operating systems, and network infrastructure.

Processes: Initially, the project management was done by PwC but later ConsCo managers became more involved with the project management. ConsCo went in mainly with an implementation methodology developed by PwC. The analysis phase led to the discovery of problems with the existing process that could be addressed by the SAP-enabled process. For instance, the accounts receivables (A/R) process at ConsCo consisted of divisional invoices, a centralized A/R department and a large number of deductions. ConsCo did not perceive the implementation as an IT project and technology was not a driving or controlling force. It was perceived to be a business project, albeit the largest one ever done. The intent was to bring in internal functional experts and give them SAP training and link them up with PwC consultants with more generic business knowledge. The project teams stepped through the business processes in a prototyping mode setting up configuration structures as appropriate. In case of problems that could not be resolved internally, ConsCo relied on PwC's network or the SAP help-desk. In case of a gap between required functionality and SAP offering, 'system mods' were used in some cases but there was recognition of issues related to maintainability and handling SAP upgrades.

During the implementation, ConsCo relied heavily on its internal staff who had gained a lot of process knowledge over the years, particularly through the implementation of a new A/R system 2 years earlier. They had also developed process models and conducted

internal surveys. The A/R team got exposure to SAP through formal classes. ConsCo did not use the SAP reference processes extensively in any part of the implementation. They relied on the collaboration of consultants from PWC who knew how A/R worked and ConsCo people who knew their business processes and what they wanted to achieve. It was found that not everything they wanted was supported by SAP. During the implementation some new workflows were implemented and were ultimately adopted by SAP as part of the core functionality.

Communication and information exchange: Initially each functional area was doing its own work with little acknowledgement of inter-relationships, but over time there was an appreciation of the integration possibilities. Integrator roles also emerged – experts who would provide their expertise to other functional areas to ease integration issues.

Evaluation of the project Overall, ConsCo's ERP project met with a mixed bag of evaluations. It was a big success for manufacturing, and on the whole 'good' for sales while a '50–50' for finance. For some of the groups it was a setback because they had been left to deal with multiple systems to work with. It was felt that had the project been fully completed it might have been a different story. The original proposal for the project had been made on a 'need-to-do' basis rather than detailed ROI projections and, hence, no formal payback analysis was conducted. The whole project was originally estimated to take 18 months. In retrospect it was hugely under-estimated. The project took 4–5 times longer than expected. The reasons for the increased time and effort were the complexity and dynamic nature of the organization and the unwillingness on the part of business units to change their processes. Also, because of the autonomy enjoyed by the business units, it was difficult to make quick decisions, as they were always subject to review by different bosses. Several times, the project groups banged heads and decisions even went up to the chairman for resolution. ConsCo had to deal with some of the limitations of SAP. For example, putting multiple legal entities onto one invoice was a difficult issue. Also, there were areas where ConsCo practices differed from SAP design. While ConsCo had separate accounts receivables for different lines of business, SAP had one pot that made it difficult to break out. ConsCo eventually learnt how to break down functional silos. With the earlier information systems, each functional area was accustomed to doing what it wanted. Work was 'thrown over' the functional walls and manipulated as needed so that the area got used to doing what it wanted. Thus the organization got resistant to changing processes. With the integrated nature of SAP, and its demands to change, this resistance was now breaking down to some extent.

Overall coordination pattern: Organization faced with high coordination needs due to fairly autonomous divisions. Cross-functional teams helped in coordina-

tion but could not entirely overcome concentration of authority at higher levels.

Case 3: UtilityCo – following the pre-configured template

UtilityCo began preparing for the implementation of an ERP system in 1996. Its legacy systems consisted of a Storeroom Inventory System and a General Ledger (Accounts Payable) System, both of which were still in use beyond their 10–15-year life expectancy. In the same year, UtilityCo underwent a 'Business System Replacement Project' to address the Y2K concerns and deregulation issues. At this time, UtilityCo identified the primary drivers for an ERP system. The implementation at UtilityCo was supported by an implementation partner with significant experience from past implementations. The company followed a very structured process of re-engineering their processes and preparing the organization for change.

Situational factors

Size and complexity: UtilityCo is a non-profit, city-owned electric utility. Compared with companies studied above, the size of the company is smaller, and the organizational structure is much less complicated as it only employs 421 people. The scope of the project was quite broad with the most processes included – Financial accounting and control (FI/CO) and Materials Management (MM) in the first phase, followed by Plant Maintenance (PM), Project System (PS), Human Resources (HR), Asset Accounting (AA) among the functionality for the second phase.

Inter-dependency and functional departmentalism: Given its relatively smaller size UtilityCo was not highly departmentalized.

Uncertainty: The level of uncertainty faced by the organization was relatively modest. Further, top-down support was perceived to be crucial to the project by the project manager. The CEO also realized that the implementation of SAP was the most important project in the organization's history. He intervened to change the corporate culture by altering some of the organization's rules and going to the board of directors to get a special bonus program approved for the project team.

Organizing elements

Organization structure: The project team consisted of one internal accountant and one external accountant each with a consultant, an internal materials manager with a consultant, and one person from each of the functional units, all with their own consultants. Included in the project team as well were two third-party ABAP programmers with one consultant between them. Team members were invited to join the team based upon knowledge of their area and the ability to make decisions.

Organization processes: The ability of the project team to make decisions during the preparation stage played a vital role in order to keep the project on schedule. Prototyping solutions and configuration options were

handled internally within the empowered team. Swift and knowledgeable problem resolution allowed top management to keep focused on day-to-day business activities. The team members came in with no knowledge of a SAP system but gained valuable technical knowledge from SAP training school. Consultants were used for their product knowledge and for prototype creation, with one major task being the construction of a 'business blueprint' of UtilityCo's organization. Function representatives were paired with their own consultant, aiding knowledge transfer. UtilityCo's approach to SAP implementation led them to simply adjust their business to SAP's 'Best Business Practices' included in the SAP reference model. The company believed it would be easier and cheaper to reengineer to fit SAP and there were no major system modifications during implementation.

Communication and information exchange: The project center was located off-site to deter distractions and its atmosphere was designed to facilitate teamwork. The center was made up of separate cubicles for each functional area all of which were connected to a WAN to allow team members to work from any of UtilityCo's facilities. Separating cubicles for each area took a little bit away from the integrated perspective needed for an ERP implementation. The implementation was expected to have benefited more from a situation in which members were proximally located. For instance, situations arose where one functional area member overheard another area discussing information requirements and realized that this involved their area and two others. If supported, close proximity would have allowed for greater understanding of dependencies to be developed.

Evaluation Overall, UtilityCo's implementation of SAP was viewed as a success. Actual project cost was just under the budget, including software, licensing, hardware, consultant fees, and training. UtilityCo underwent numerous types of internal and external evaluations. Prior to implementation, a Quality Assurance (QA) review was conducted by SAP. In order to assess the level of understanding and effectiveness following end-user training classes, evaluations were given to the employees in attendance. Post-implementation, UtilityCo realized a few 'surprises' such as the limitation of the reporting module, despite great effort invested in that area.

Overall coordination pattern: Moderate coordination needs due to stable environment. Empowered project teams ensured successful coordination.

Case 4: GlobalFinCo – executive commitment to the fore

GlobalFinCo instituted the 'Systems Renewal Initiative' to move from legacy systems to integrated commercial platforms. This was a centerpiece of the strategy for change in the organization and was adopted by a group of committed top managers. It was also made a part of the 'Strategic Compact' struck between management and the board of executive directors. Thus, while the ERP project

was spearheaded by the IT function the business functions were actively involved through formal mechanisms.

Situational factors

Size and complexity: GlobalFinCo is a global financial institution with a staff of about 10,000 spread across the world. In the 1990s, GlobalFinCo's president created a vision for the organization to become a knowledge organization – a world-class repository of experience and best practices that would benefit all entities that did business with it. However, by late 1996 it became clear that the IT systems were not in place to support this vision. The core applications supporting GlobalFinCo's operational business processes suffered from duplication and fragmentation of data. Also, there were complex data dependencies between systems that few people could understand and keep track of. In preparing the annual report for 1996, for example, the financial database that was used to extract the data was later found to be 'out-of-sync' with the master database from which it was periodically refreshed. This required the master databases needing to be changed to agree with the annual report even though the changes were erroneous. While the weaknesses of GlobalFinCo's systems were known, they were masked with compensatory administrative procedures such as vetting and reconciliation by support staff. Another issue was the proliferation of application systems and databases because of the decentralization of application development. At one point in time there were more than 65 application systems and more than 100 databases leading to a 'spaghetti' architecture that was not responsive to business needs.

Inter-dependency and functional departmentalism: GlobalFinCo is organized as a matrix organization with function as well as region-based departments. In February 1997, the organization's centralized information management function set out to reform GlobalFinCo's administrative and operational systems. A non-technical person was appointed to head the technology group – this helped to build bridges between technical personnel and business users, and also corrected the 'technology for technology's sake' syndrome that had become prevalent. In mid-1997 this person was designated the CIO for the organization. One of the other changes to the IT function was to set up an enterprise-wide network for IT personnel that would allow them to be affiliated with the centralized IT function and also with the operational functions. This allowed for cross-functional coordination on IT issues.

Uncertainty: GlobalFinCo employees had a dominant belief that the organization was unique and thus required customized solutions, creating opposition to standardized solutions. The new leadership emphasized a move towards standardized technology solutions and the courage to rethink and change processes. The final timeline for the implementation of the chosen ERP package was also influenced by the need to get the organization ready for the Y2K issue.

Organizing elements *Organization structure:* The implementation was done over a 15-month timeline. The core project team consisted of GlobalFinCo's internal IT staff and consultants from the prime contractor and grew to about 60 people. GlobalFinCo also used offshore development with staff from the prime-contractor's offshore location contributing about 15 man-years of development time. In order to make sure that the business side was actively involved in the project, the project organization included business sponsors that took ownership of the new processes. In all, 20–30 business users were actively involved at various stages in making decisions about the functionality. The project structures put in place for the implementation have been found to be the stepping stone for future coordination as well.

Organization processes: GlobalFinCo chose to go with the SAP enterprise solution and chose Lotus Notes to supplement it. While the SAP package tended to take a structured data perspective GlobalFinCo's processes were more document-centric and therefore needed the Notes capability. In most cases, the organization chose to go with the SAP designed processes with some modifications through 'bolt-on' code. However, there were still as many as 90 major process changes. The treasury function, with a more convoluted and a process specific to the organization, was left out from the scope of the implementation. One of the key elements of the implementation was the effort expended in training and dealing with change management issues – about 20% of the budget was also earmarked for training – but in retrospect, even that was deemed too low.

Communication and information exchange: At GlobalFinCo information exchange across functional units was accomplished by having business sponsors continually updated on major developments. Consensus was built on important design issues. A complicating factor was the need for addressing the concerns of overseas locations. This was done by taking advantage of existing organization structures and relationships that provided an avenue for overseas locations to be represented.

Evaluation Overall, the implementation was judged to be a success and met its time and resource goals but there was some user dissatisfaction with the system interface. There were also some initial teething problems. In the first month after implementation, the major issues were related to user security profiles and access issues. In the next month, there was a major backlog of transactions as users were adjusting to using the system. One particular module related to travel was particularly drawing most of the helpdesk traffic. About 3 months after the system went live, most of the issues cropping up were limited to employees asking for specific reporting functionality or complaints about a process requiring two or three screen traversals whereas it had been one screen interface in the past. One of the major benefits of the new ERP system – the integration between departments – also introduced problematic issues – the need for putting appropriate

controls that had not been a requirement with standalone systems. User comments attested to this problem – 'Any one can charge the projects for which I am responsible and I cannot prevent it. I have to spend my resources chasing errors...introduced by others'.

Overall coordination pattern: IT function spearheaded coordination efforts while functional units kept engaged and responsible through process ownership

Summary of cases Table 4 provides a summary of the conditions for the different ERP implementations that were studied. In line with the contingency perspective presented earlier we particularly emphasize the conditions that lead to enhanced coordination needs and the organization's ability to match it with the required level of information processing.

A comparative analysis of functional coordination for each organization suggests that organizing elements identified earlier are significant in affecting project outcomes. For example, a balanced empowered team of people, representing and communicating with their business functions can be very effective, as UtilityCo's experience demonstrated. Strong top management leadership was a major success factor for UtilityCo, but could not overcome the functional conflict at TeleEquipCo. Through these rich case studies, we outline important lessons related to the relationship between situational factors and organizing elements that are enacted by these enterprises.

Discussion

A central issue that affects functional coordination requirements in ERP implementations is how to address the gap in functionality between that required by the organization and that provided by the package. Organizations have to choose among: adapting to the new functionality, living with the shortfall, instituting work-arounds, or customizing the package (Soh *et al.*, 2000). Given that the scope of ERP systems spans multiple organizational units, coordination across functional areas is important to ensure that the system is effectively implemented.

Before discussing the findings of this study, we would like to point to some limitations of our research methodology, which require that the results be interpreted with appropriate caution. First, our understanding of the four ERP projects is largely based on qualitative insights provided by a small set of respondents. While confirmation by multiple respondents helps to increase our confidence in the validity of the findings, personal biases in the construction of the narratives cannot be ruled out. Second, due to the intensive nature of our data collection and analysis, our study is limited to four large organizations. Thus, the generalization of our findings to other contexts must be done very cautiously, particularly as past studies (e.g., Buonanno *et al.*, 2005) indicate significant differences between small/medium and large organizations on their ERP implementation practices.

Table 4 Summary of case studies of ERP implementations

	<i>TelEquipCo</i>	<i>ConsCo</i>	<i>UtilityCo</i>	<i>GlobalFinCo</i>
Situational factors leading to coordination needs	Very high Management model is complex; Independent functional units; Complicated management model of 'constructive contention'; Medium to large project scope; Contentious ERP choice – lack of business buy-in;	Very high Large organization with dynamic environment; Multiple divisions with relatively high autonomy; Large project scope; Complex legal entity structure compounded by spin-off	Moderate Small organization with stable environment; Large project scope; Major reengineering needed; Pressures for industry deregulation	Moderate Critical issue was coordinating with overseas locations; Religion wars kept to a minimum with executive mandate to relentlessly standardize
Organizing elements facilitation coordination	Practice of getting people from the business units, training them and then sending them back to the business units; Follow reference models closely	Multi-disciplinary teams representing functions; Quick learning culture to adjust: • From part-time to full-time members • More reference model based approach from customization • Emerging 'integrator' roles	Project team isolated from distractions; Strong top-down support; Cultural changes made; Strong team building	Each project has formal business sponsor; Funding for front-line staff so that they could serve as regular advisors; Rigorous contract management and well-defined methodology
Emergent characteristics of functional coordination	Conflict Functions made independent decision on consultant support leading to different methodologies; After breakdown, conscious attempt to enforce standards; Persistent 'Bias' to do things functionally	Resistance Goal of single invoice could not be realized – pharmaceutical and consumer divisions could not agree as different industry payment terms; Difficult to make quick decisions, as subject to review by different bosses	Team based Team knowledgeable and highly empowered; Teams represented each function and own consultants; Equipped with communication tools (cellphones, ebuletin boards, etc.)	IT Driven Highly consultative process with business case construction to prioritize investments; Many managers in regions and functional groups involved in planning activities; Business people 'owned' the process, IT staff implemented
Implementation outcome	Failure to realize benefits from implementation; Breakdown and then re-initiation	Mixed Success – success for manufacturing, not for sales, financials; Project took 4–5 times longer than expected	Success with a few surprises – financial and human costs; System navigation and reporting concerns	Success implemented on time under constrained budget; End-user satisfaction issues

Thirdly, in focusing on a small set of coordination related factors, we may have omitted controlling for other factors that may have played a significant role in impacting project outcomes. Lastly, this paper reports on ERP projects mostly executed in the late 1990s. There have naturally been changes in the basic information technologies as well as in ERP package features subsequent to this. One such change has been the convergence of enterprise systems and enterprise application integration (EAI) approaches (a recent special issue of *EJIS* addresses these issues). However, we believe that cross-functional coordination issues are of enduring interest to researchers and practitioners. While the technical issues in ERP projects may change very rapidly, organizing for coordination is expected to be important for different generations of technology. Our research questions, orienting framework, and takeaways are general enough to apply to contemporary and future projects.

Our findings present some important insights. First, the organizing elements significantly interact with situational factors. Organizational size and complexity impede functional coordination as shown in the case of TeleEquipCo. One of the major barriers to a successful implementation at ConsCo included the size and complexity of the implementation, functional silos, and the autonomy of individual businesses. ConsCo's quick movement toward the more integrator roles and more reference model-based approach is closely related to a flexible organizational culture. Second, not only do those organizing elements interact with situational factors, but also there are interactions among the organizing elements. One of the reasons for success at UtilityCo was the fact that team members could make decisions on their own, and it could not happen without strong top-management support. The open communication among team members in the ConsCo is also related to constituents and location of project teams.

Based on an analysis of our case studies, we find that cross-functional coordination in ERP implementations may be advanced through three alternate means:

(a) Lean coordination – structured inter-dependencies (standardization)

The first pathway to aid cross-functional coordination is based on reducing the coordination requirements by following a vanilla² approach to implementation using reference models. This approach allows for functional dependencies and interfaces to be easily discovered and managed by referring to standardized implementations. Standardization is an effective way to achieve coordination but has been neglected in previous ERP studies. Standardization can also create common frames of reference. At ConsCo, for example, the repository of

reference processes created through the PwC tool was also used as an effective communication tool inside project teams. One of the things that ConsCo thinks it could have done differently, in retrospect, was to have gone for an even more 'vanilla' implementation, that is, made fewer customizations to the reference models that are provided by the ERP vendors.

It is, however, important for organizations to critically evaluate adherence to standards as institutional pressures on organizations may lead to unquestioned adoption of 'misfit' business processes (Gosain, 2004). Best practices set forth by ERP vendors may themselves be socially constructed and fuel tensions among groups of actors involved in implementations (Wagner *et al.*, 2004).

(b) Rich coordination – manage inter-dependencies (mutual adjustment)

The second pathway relies on management of inter-dependencies between functions, through appropriate project team composition and fostering of a collaborative culture that stresses integrated processes. This requires:

- Project team members should be very knowledgeable about their functional business process, be empowered, and be able to influence the decision making in their functional area. This was the case at UtilityCo.
- Project teams need a channel through which conflict between functional areas can be quickly resolved (e.g. through regular meetings with the steering committee).
- Nurturing an organizational culture in which cross-functional coordination is achievable and efficient. Top management commitment as well as team-building exercises, through which team members recognize the importance of the project for the enterprise ahead of their functional interest, appear to be important as well.

Based on the case studies, we have generated two functional coordination models relying on mutual adjustment. One emphasizes coordination at the level of the project teams and the other one at the level of the functional units.

Mutual adjustment – empowered teams This model (see Figure 2) is based on the case of UtilityCo. In this organization most of communication and coordination activity occurs within the project team. Once consensus is achieved, the functional units are informed and accept it. Critical success factors for this model are team members who are both knowledgeable and empowered. A pitfall of this model is that the team might not completely address functional unit needs due to increasing identification and engagement with the team, or lose track of changes occurring at the functional level. Limitations in reporting functionality at UtilityCo were ascribed to this.

²A vanilla (or plain vanilla) approach refers to implementations of pre-configured packages with as little customization as possible. ERP vendors provide industry-specific templates that are designed to minimize the need for customization.

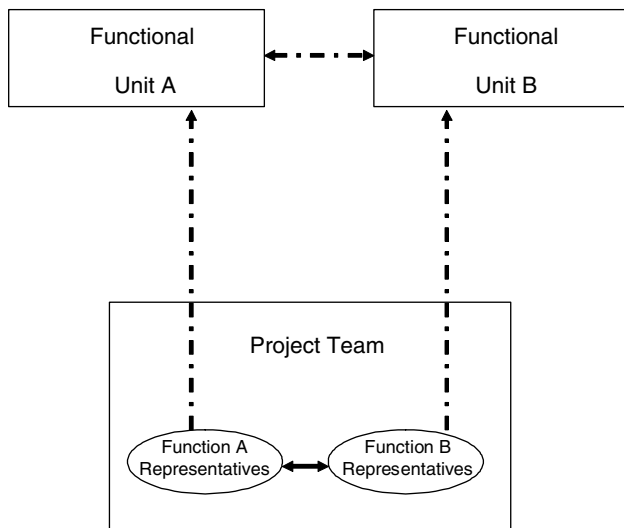


Figure 2 Mutual adjustment with empowered teams.

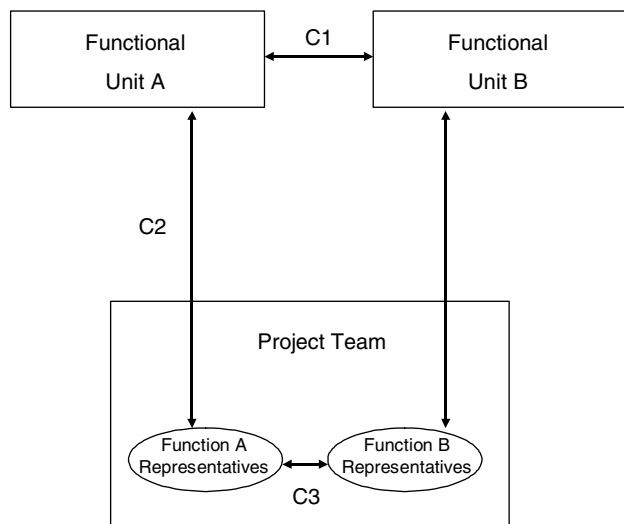


Figure 3 Mutual adjustment at functional units.

Mutual adjustment – functional units The second mutual-adjustment based model (Figure 3) draws on TelequipCo and ConsCo cases. In this model, coordination is initiated first by communication inside the project team (C3), followed by escalation to each functional unit (C2), and final coordination efforts between functional units (C1). In the case of TeleEquipCo, there was a communication gap between functional units (C1), and even inside the project team (C3). In ConsCo's project, while they had relatively open negotiations among functional units (C1), often times this lateral coordination required time to resolve and resulted in project delays.

The critical success factors for this model of coordination are quick conflict resolution in C1, and it can be achieved through top-management leadership or structured conflict resolution mechanisms. Sometimes, as was

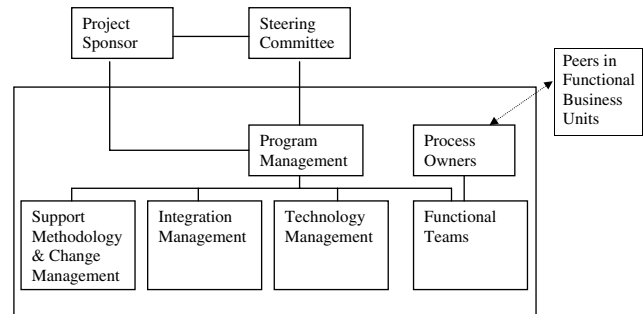


Figure 4 Supporting effective mutual adjustment.

the case for ConsCo, this may be occur naturally through informal mechanisms:

Initially each functional area was doing its own work with little acknowledgement of interrelationships, but over time there was an appreciation of the integration effect of the system. A finance person would, for instance, be asked by a manufacturing person on what account to specify for a transaction. Also, some of the more curious people felt like 'seeing what the other side was doing'. Later in the project, the silo effects progressively reduced as there was a greater emphasis on cross-functional assignment.... [ConsCo Project manager]

An example of a project management structure that could support effective mutual adjustment is shown in Figure 4.

(c) Mediated coordination (direct supervision) The third pathway to coordination relies on a mediating entity that commands clout in the organization to impose an edict when conflict emerges. The project manager at TelEquipCo strongly pointed to the need for a benevolent dictator that could make decisions for the organization eliminating dissipative conflict. Without such a role, TelEquipCo spent much of its time debating the choice of the specific ERP software and in the end could not get buy-in from the functional areas to implement an integrated solution.

A case where mediated coordination succeeded is the GlobalFinCo implementation. At the very outset of the project the president of the company charged a senior 'non-techie' person to take charge of the IT solutions group. Further, the top executives signed a compact agreeing to the basic blueprint of the implementation and empowering the IT group to spearhead the implementation. The IT group also secured a mandate from the top executives to relentlessly standardize processes and consolidate and integrate the infrastructure. The functional areas worked with the IT group by creating liaison roles to communicate their viewpoint, but the head of the IT group took on the task of forging consensus or, in some cases, taking quick common-sense decisions on issues that had the potential to become foci for lengthy debate. The communication process was very

Table 5 Prescriptions for different cross-functional coordination patterns

Coordination through...	Key imperatives
Standardization	Structure: Cross-functional teams planning to develop baseline mental models, frameworks, and language Processes: Organization should be willing to give up idiosyncratic processes or should operate using common business practices Information exchange: Upfront planning on formal methodologies and understanding reference models
Mutual adjustment – empowered teams	Structure: Empowered, knowledgeable project teams; flexible and non-hierarchical structures Processes: Delegation of responsibility to team liaisons Information exchange: Organization culture supporting information sharing and facilitating quick decision making
Mutual adjustment – functional units	Structure: Balanced team composition Processes: Quick reporting to functional areas Information exchange: Rich communication across functional areas
Mediation	Structure: Competent and responsive authority figures Processes: Clear process ownership and implementation responsibilities Information exchange: Vision of ‘life after implementation’ developed and shared

important for the success of the implementation. Apart from townhall type of meetings, the IT executives engaged their business counterparts through sidebar conversations across lunch or coffee to ensure buy-in. One of the interesting things that resulted from this implementation was a change in the perceptions of the IT function – from a provider of IT, it was now starting to be perceived as a provider of information-based solutions.

Table 5 presents the situations in which each pattern of functional coordination may be best suited. As is evident, the coordination models are not alternatives but may be employed in conjunction. It appears, however, that organizations tend to use more of one flavor. In our case studies with some amount of coordination success – ConsCo predominantly chose the mutual adjustment approach, UtilityCo went with standardization and GlobalFinCo went for mediation. TelEquipCo could not secure the coordination to be a success – it may be partly ascribed to the lack of stable leadership as the company went through three different leaders for the project over time.

Our findings show that organizations may start with similar coordination needs but take different pathways to coordination to ultimately achieve similar coordination outcomes. UtilityCo used mutual adjustment while GlobalFinCo used mediation but both achieved similar success. The concept of *equifinality* has been proposed by researchers in organizational design to augment the contingency perspective. It suggests a functional equivalence-based analysis where the research goal is less oriented to the specification of a particular fit-performance relationship and more directed at specifying which fit-performance relationships occur in which contexts, described as ‘second-order contingency theory’ (Gresov & Drazin, 1997). In line with this perspective, our study has provided illustrations of the specific coordination patterns that result in different contexts.

This research has tried to study ERP implementations by employing the case study based theory development focused on understanding the nuances of functional coordination. There is a need to test if the emergent modes of coordination outlined in this study do reflect the approaches followed in the larger population of ERP implementations. While this research has looked at ERP implementations in large enterprises with multiple business units, where the needs for cross-functional coordination are exacerbated, future research can examine this issue in the context of medium and smaller-sized enterprises.

It is important to add a cautionary note in relation to the organizing elements corresponding to different coordination patterns prescribed above. While organizations have some leeway in making interventions to foster a specific type of coordination, they also need to be conscious of ‘drift’ beyond the intended impact³. Drawing upon actor-network theory, Ciborra & Hanseth (1998) suggest that infrastructures may be understood as formative contexts, where the tension between technology inscriptions and the process of translation causes the installed base to ‘drift’. The way infrastructures are implemented and used may not correspond to the designer’s original plans and visions. Thus, the emergent patterns of coordination may not be those intended by the stakeholders.

Conclusion

This paper has explored cross-functional coordination in ERP implementation settings. An emergent process perspective explored through case studies enables us to richly characterize the different patterns of cross-functional coordination. Effective coordination may occur (1) through

³We thank an anonymous reviewer for pointing us to this literature and Claudio Ciborra’s seminal insights.

structuring and standards to minimize inter-dependencies (lean coordination), or (2) through intense and responsive communication that manages those inter-dependencies (rich coordination), or (3) through mediation by an empowered entity capable of making resource allocations and resolving conflict. In the lean coordination case, the use of formal process reference models and adherence to rigid specifications pre-built into the package ensures that different functional areas stay coordinated. In the rich coordination case, the organization does not enforce formal structure, and coordination is enabled by rich communication and collaborative process mechanisms.

The main contribution of this study is in showing that contingency conditions in organizations need to be

carefully assessed before making interventions to facilitate technology implementations. For researchers, this research suggests that an emergent process perspective that takes contextual process factors into account may help in understanding the failure of 'conventional' factors (Sarker & Lee, 2000) to account for ERP implementation success. For practitioners, it highlights the importance of coordination among functional areas to facilitate the implementation of complex enterprise technologies. Future research can validate the patterns of cross-functional coordination across a larger sample of organizations, and also investigate how the patterns of coordination evolve over time as the technology gets assimilated in the organization.

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Appendix

The interview protocol used for this study employed a structured questionnaire with the following key areas. The stages are identified based on Markus & Tanis (2000).

Choice stage

- What was the basic motivation in going in for an ERP system?
- Problems with earlier system or processes
- Perceived opportunity from new system and processes – what did they want to achieve
- Competitor moves
- Industry trends
- What kind of analysis was performed in choosing the ERP package?

Planning stage

- What kind of planning was done before implementation?
- What kind of capability assessment was done?
- What were the key competencies that company had to source from outside?
- What kind of customization or process change was envisaged?
- Perception of organization knowledge and resources to handle implementation

Preparation stage

- How were the existing processes analyzed and new ones designed?
- Role of business process reference models, documentation and flow charting tools
- Project organization and structures
- Any information systems to assist the implementation

Implementation stage

- Steps in implementation process
- Key roles in implementation
- Extent of their customization effort? (i.e. what are things they have to change from the reference model to accommodate into their organization)
- What kind of workarounds
- What kind of help/support mechanisms were instituted
- Was there evidence of informal knowledge exchange mechanisms (communities of practice)
- Was there evidence of learning from outside the organization (formal/informal)

Evaluation

- Evaluation of current implemented process (success, problems?)
- Financial, operational, customer satisfaction and other indicators before and after
- Misfit types
- Data (format, relationships)
- Function (access, control, operational)
- Outputs (presentation, content)
- What kind of knowledge was gained during the implementation about existing business processes and new processes?
- How much of tacit knowledge was captured explicitly
- Are there deviations from designed process (users use differently)
- In retrospect, would they have managed process differently

Other information sought

- Examples of legacy and designed process segments
 - Performance benchmarks
 - Internal satisfaction, efficiency surveys
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